

Section 13.1 Functions of Two or More Variables

Topics: (1) Definition of Functions

(2) Finding Domains

(3) Graphing Functions of Two Variables

(4) Level Curves

Recall: Functions of Single Variable $y = f(x)$, $x = f(y)$

$y = x^2$: Input = y . Then it's not a function of y , because if $y = 4$, then $x = \pm 2$.

But $y = x^2$: Input = x . Then it's a function of x . So, $y = f(x) = x^2$.

Note: In a function, output cannot have an even power.

Recall: Definition: For any real number x , $\sqrt{x^2} = |x|$. For example, $\sqrt{16} = 4$

Writing a function from a non-function:

$$y = x^2 \Rightarrow x = \pm\sqrt{y}$$

Note that $\sqrt{9} = 3$ and $-\sqrt{9} = -3$

Positive Square Root: $x = \sqrt{y}$. Here $x \geq 0$. $D = [0, \infty)$ and $R = [0, \infty)$

Negative Square Root: $x = -\sqrt{y}$. Here $x \leq 0$. $D = [0, \infty)$ and $R = (-\infty, 0]$

Observation:

Complex Number:

$\sqrt{-1} = i$, that is, $i^2 = -1 \rightarrow$ imaginary identity, $z = a + bi \rightarrow$ imaginary/complex number.

Note that $\sqrt{-1}$, $\sqrt[4]{-1}$, $\sqrt[6]{-1}$, $\sqrt[even]{-1} \dots$ are also imaginary number.

Radical: The n^{th} root of x is denoted by $\sqrt[n]{x}$ and is called the radical.

- x is any real number
- n is a positive integer, and n is called the index of the radical.
- If n is even and if x is a negative number, then $\sqrt[n]{x}$ is an imaginary number.
- $\sqrt[n]{\text{negative}}$ is an imaginary number.

A **function** is a relation between a set of inputs and a set of permissible outputs with the property that **each input is related to exactly one output**.

$[y = f(x) = x^2, \text{ If } x = 1, \text{ then } y = 1 \text{ and if } x = -1, \text{ then } y = 1]$, this is a function, but not 1-to-1.

Definition: Let X and Y be two nonempty sets. A **function** from X into Y is a relation that associates with each element of X exactly one element of Y

Note: For a function, no input has more than one output. The **domain** of a function is the set of all inputs; the **range** is the set of all outputs.

Undefined: $\frac{\text{non-zero}}{0}$ and complex: $\sqrt[n]{-1}$. [$\frac{0}{0} = \text{indeterminant}$, $\frac{0}{3} = 0$]

Odd roots are not a problem, because $\sqrt[n]{-1} = (-1)^{\text{odd}} = -\sqrt[n]{1} = -1$.

For example, $(-1)^7 = -1$. Hence, $\sqrt[7]{-1} = -1$ and $\sqrt[3]{-8} = -\sqrt[3]{8} = -2$.

Note: If $x^{\text{even}} < 0$, then x is a complex number.

Domain: Collection of all inputs so that the outputs are **defined** and **real**.

Domain of $y = \sqrt[n]{x+1} = \frac{\sqrt[n]{x+1}}{1}$. For the domain, $x+1 \geq 0$. $D = [-1, \infty)$

Domain of $y = \sqrt[n]{x+1} = \frac{\sqrt[n]{x+1}}{1}$. $D = (-\infty, \infty)$

NOTATION AND TERMINOLOGY

There are many familiar formulas in which a given variable depends on two or more other variables. For example, the area A of a triangle depends on the base length b and height h by the formula

$$A = \frac{1}{2}bh; \rightarrow A = f(b, h), \text{ input} = (b, h)$$

the volume V of a rectangular box depends on the length l , the width w , and the height h by the formula

$$V = lwh; \quad V = f(l, w, h), \quad \text{input} = (l, w, h)$$

and the arithmetic average $\bar{x}$ of n real numbers, $x_1, x_2, \dots, x_n$, depends on those numbers by the formula

$$\bar{x} = \frac{1}{n} (x_1 + x_2 + \dots + x_n) \rightarrow \bar{x} = f(x_1, x_2, \dots, x_n), \text{ input} = (x_1, x_2, \dots, x_n)$$

Thus, we say that

A is a function of the two variables b and h ;

V is a function of the three variables l , w , and h ;

$\bar{x}$ is a function of n variables $x_1, x_2, \dots, x_n$.

The terminology and notation for functions of two or more variables is similar to that for functions of one variable. For example, the expression

$$z = f(x, y) \quad [\text{For } (x, y, z), \text{ here } z \text{ is the output}]$$

means that **z is a function of x and y** in the sense that a unique value of the dependent variable z is determined by specifying values for the independent variables x and y .

Definition: A **function f of two variables**, x and y , is a **rule** that assigns a unique real number $z = f(x, y)$ to each point (x, y) in some set D in the xy -plane.

Graph $f = \{(x, y, z): z = f(x, y) \text{ for } (x, y) \in D\}$

Example: $f(x, y) = e^x \cos y$, $f(x, y) = \cos(x^2 y)$, $z = f(x, y) = 1$

Note: $y = f(x) = 1$ is a horizontal line. That means, $z = f(x, y) = 1$ is a horizontal plane.

Note: Given that $z = f(x, y) = e^x \cos y + 1$.

Projection on the xy -plane $z = 0$: $e^x \cos y + 1 = 0$, that is, $y = \cos^{-1}(-e^{-x})$

Similarly,

$$w = f(x, y, z)$$

expresses w as a function of x, y , and z .

Definition: A **function f of three variables**, x, y , and z , is a **rule** that assigns a unique real number $w = f(x, y, z)$ to each point (x, y, z) in some set D in three dimensional space.

Example: $f(x, y, z) = x^2 \cos y \sin z$, : $f(x, y, z) = x^2 \cos(xyz^3) \sin(yz)$,

$$w = f(x, y, z) = 1$$

Domain

Unless specifically stated otherwise, the domain of a function of several variables is the set of all values of the variables for which the given expression is defined (It gives real values).

Recall: Domain of a Function of a single variable $y = f(x)$

Domain of a function f is the largest set of real numbers for which the value $f(x)$ is a real number.

Finding the Domain of a Function Defined by an Equation:

1. Start with the domain as the set of all real numbers.
2. If the equation has a denominator, exclude any numbers that give a zero denominator.

3. If the equation has a radical of even index, exclude any numbers that cause the expression inside the radical to be negative. [$\sqrt{-1}$, $\sqrt[4]{-1}$, $\sqrt[6]{-1}$, ... Not real]

Good: $\sqrt[3]{-1} = -1$, $\sqrt[3]{-8} = -2$, $\sqrt[odd]{-a} = -\sqrt[odd]{a}$ where $a > 0$, Always real. Also, $\sqrt[n]{0} = 0^{\frac{1}{n}} = 0$.

Recall: $\sqrt[n]{x}$ is a radical, called n -th root. The number n is a positive integer, called the index.

Note: $\frac{1}{0}$ is undefined. $\sqrt{-1} = i$, $\sqrt[even]{-1}$ is NOT real.

Note that $x^2 \geq 0$ for any real number x . If $\sqrt[4]{-1} = a \Rightarrow a^4 = -1$, which is not true for any real number a .

Example: Find the domain of the following functions:

1) a) $f(x) = x^2 + 4$. **Answer:** Domain = all real numbers

$$b) (1) f(x) = \frac{x^2+1}{x^2-5x+6} = \frac{x^2+1}{(x-2)(x-3)}$$

Answer: Domain = $\{x \in \mathbb{R} : x \neq 2, 3\} = (-\infty, 2) \cup (2, 3) \cup (3, \infty)$

[Note that if $x \in (2, 3)$ means $2 < x < 3$; if $x \in [2, 3]$ means $2 \leq x \leq 3$; and if $x \in [2, 3)$ means $2 \leq x < 3$]



(2) $f(x) = \frac{x^2+1}{x^2+6}$; $D =$ All real numbers.

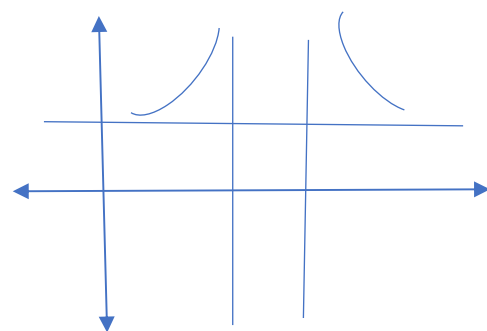
$$a) f(x) = \sqrt{2x+4} = \frac{\sqrt{2x+4}}{1}$$

Answer: For the domain, set $2x + 4 \geq 0 \Rightarrow 2x \geq -4$.

Domain = $\{x : 2x + 4 \geq 0\} = \{x : x \geq -2\} = [-2, \infty)$

(2) b) (i) $f(x) = \sqrt[5]{2x+4}$; $D =$ All real numbers.

Note that $\frac{0}{1} = 0$.



b) (ii) $f(x) = \frac{5x-3}{\sqrt{3x-7}}$

Answer: Domain = $\{x : 3x - 7 > 0\} = \left\{x : x > \frac{7}{3}\right\} = \left(\frac{7}{3}, \infty\right)$.

Note: $\sqrt[n]{0} = 0$.

Note that $3x - 7 \neq 0$. If $3x - 7 = 0$, then $\sqrt{3x - 7} = 0$. Hence, $\frac{5x-3}{\sqrt{3x-7}}$ is undefined.

So, for the domain, we want $3x - 7 > 0 \Rightarrow x > \frac{7}{3}$.

$f(x) = \frac{5x-3}{\sqrt{3x-7}}$; Domain = $\left(\frac{7}{3}, \infty\right)$.

(ii) $f(x) = \frac{5x-3}{\sqrt[11]{3x-7}}$, $D = \left\{x \in \mathbb{R} : x \neq \frac{7}{3}\right\} = \left(-\infty, \frac{7}{3}\right) \cup \left(\frac{7}{3}, \infty\right)$

Sum, Difference, Product and Quotient of Two Functions:

If f and g are two functions, then

1. The **sum** $f + g$ is the function defined by $(f + g)(x) = f(x) + g(x)$

The domain of $f + g$ = domain of $f \cap$ domain of g .

2. The **difference** $f - g$ is the function defined by $(f - g)(x) = f(x) - g(x)$

The domain of $f - g$ = domain of $f \cap$ domain of g .

3. The **product** $f \cdot g$ is the function defined by $(f \cdot g)(x) = f(x) \cdot g(x)$

The domain of $f \cdot g$ = domain of $f \cap$ domain of g .

For Examples, (1) $f(x) = \sin x$, $g(x) = \ln x$.

Domain of $f = (-\infty, \infty)$ and Domain of $g = (0, \infty)$

Now, let $h(x) = (f \cdot g)(x) = f(x) \cdot g(x) = \sin x \cdot \ln x$, that is, $h(x) = \sin x \cdot \ln x$

Domain of $h = \text{Domain of } f \cap \text{Domain of } g = (0, \infty)$.

(2) **Example:** Given, $f(x) = \frac{x-2}{x-1}$, $g(x) = \frac{x}{x-2}$

$h(x) = (f \cdot g)(x) = f(x) \cdot g(x) = \frac{x-2}{x-1} \cdot \frac{x}{x-2}$, that is, $h(x) = \frac{x}{x-1}$; $x \neq 2$

4. The **quotient** $\frac{f}{g}$ is the function defined by $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$.

The domain of $\frac{f}{g} = \{x \mid g(x) \neq 0\} \cap \text{domain of } f \cap \text{domain of } g$.

Example: Given, $f(x) = \frac{1}{x-1}$, $g(x) = \frac{x}{x-2}$

Note that $g(x) = \frac{x}{x-2} = 0$ iff $x = 0$.

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\frac{1}{x-1}}{\frac{x}{x-2}}, x \neq 1, 2 = \frac{x-2}{x(x-1)}; x \neq 1, 2$$

The domain of $\frac{f}{g} = \{x \mid g(x) \neq 0\} \cap \text{domain of } f \cap \text{domain of } g$

$$= \{x \mid x \neq 0, 1, 2\} = (-\infty, 0) \cup (0, 1) \cup (1, 2) \cup (2, \infty)$$

.....

Note that, the domain of $y = \frac{x-2}{x(x-1)}$ is

$$= (-\infty, 0) \cup (0, 1) \cup (1, \infty)$$

➔ Note: We need to know Trigonometric, Exponential and Logarithmic functions.

Exponential Function: $y = b^x$, $b > 0$, $b \neq 1$; Inverse: $y = \log_b x$

(1) If $0 < b < 1$, then $y = b^x$ is decreasing. (2) If $b > 1$, then $y = b^x$ is increasing.

Examples: (1) Consider, $y = 2^x$, $y = \pi^x$, $y = e^x$, $D = (-\infty, \infty)$, $R = (0, \infty)$

(2) (a) Consider, $y = \pi^{\sqrt{x}}$, $D = [0, \infty)$, $R = [1, \infty)$

(b) Consider, $y = \left(\frac{1}{2}\right)^{\sqrt{x}}$, $D = [0, \infty)$, $R = (0, 1]$

(3) Consider, $y = \pi^{-\sqrt{x}} = \frac{1}{\pi^{\sqrt{x}}} = \left(\frac{1}{\pi}\right)^{\sqrt{x}}$, $D = [0, \infty)$, $R = (0, 1]$

(4) $y = \tan x = \frac{\sin x}{\cos x}$; Domain = $\{x : x \neq (2n + 1)\frac{\pi}{2} \text{ for any integer } n\}$

Note: $y = \frac{x-1}{x^2-1} = \frac{1}{x+1}$, because $\frac{x-1}{x-1} = 1, x-1 \neq 0$.

$$y = \frac{x-1}{x^2-1} = \frac{1}{x+1}, x \neq 1 = \frac{1}{x+1} \cdot \frac{x-1}{x-1}.$$

Exercises from MAT-250

Exercise: 1 Consider the function $f(x, y) = \sqrt{x+1} - \log_2(4 - x^2 - y^2)$.

(a) Evaluate $f(0, 0)$, $f(0, \sqrt{2})$, $f(3, -1)$ and $f(1, -1)$

(b) Find the Domain of the function f , and **sketch it**.

$$\text{Solution: } f(0, 0) = \sqrt{1} - \log_2 4 = 1 - \log_2 2^2 = 1 - 2 = -1,$$

$$f(0, \sqrt{2}) = 1 - 1 = 0$$

$$\text{and } f(1, -1) = \sqrt{2} - \log_2 2 = \sqrt{2} - 1$$

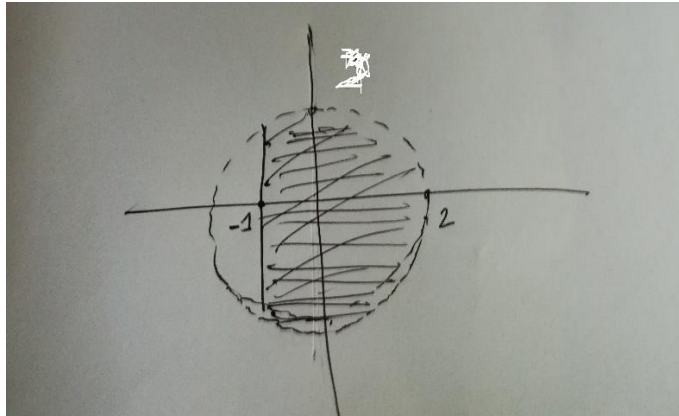
Note that $f(3, -1)$ is not defined, because -6 is not in the domain of the log function.

For the domain of $f(x, y) = \sqrt{x+1} - \log_2(4 - x^2 - y^2)$:

We need to have $x+1 \geq 0$ and $4 - x^2 - y^2 > 0$

That is, $x \geq -1$ and $x^2 + y^2 < 2^2$ [note that $x^2 + y^2 = 2^2$ is a circle]

Domain: $\{(x, y) : x \geq -1 \quad \text{and} \quad x^2 + y^2 < 2^2\}$



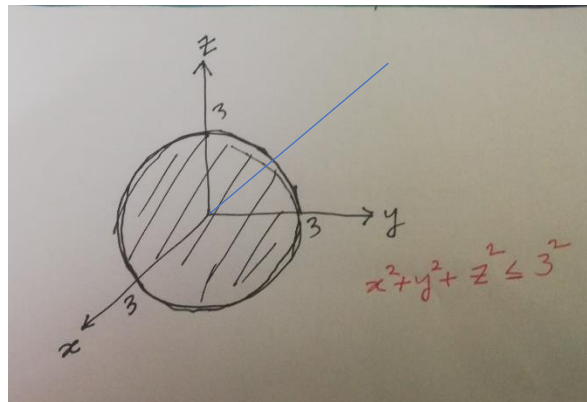
Exercise: 2A Consider the function $f(x, y, z) = \sqrt{9 - x^2 - y^2 - z^2}$

(a) Evaluate $f(0, 0, 0)$, $f(0, \sqrt{2}, 2)$ and $f(1, 3, 0)$

➔ Since $f(1, 3, 0)$ is not real, hence $(1, 3, 0)$ is not in the domain.

(b) Find the Domain of the function f and sketch it.

For the domain, set $9 - x^2 - y^2 - z^2 \geq 0 \Rightarrow x^2 + y^2 + z^2 \leq 3^2$



Exercise: 2B Consider the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2 - 4}$

Exercise: 2C Consider the function $f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2 - 4}}$

Exercise: 2D Consider the function $f(x, y, z) = \frac{2x+y^2-1}{\sqrt{4-(x^2+y^2+z^2)}}$

Definition: The domain of the function $y = (f \circ g)(x) = f(g(x))$ is

$$D = \{x : g(x) \in \text{Domain of } f\}$$

Exercise: 3 Consider the function $f(x, y) = \sin\left(\frac{x+y}{\sqrt{x^2-y}}\right)$

[note that if $z = \frac{x+y}{\sqrt{x^2-y}}$ is real, then it will be in the domain of the *sine* function]

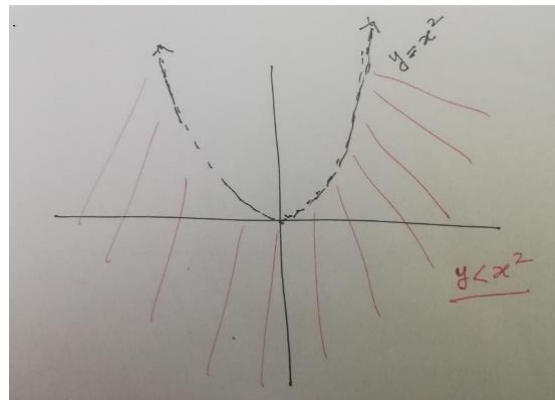
(a) Evaluate $f(0, 0)$, $f(\sqrt{2}, 0)$ and $f(1, -1)$

(b) Find the Domain of the function f .

$$\text{Domain} = \{(x, y) : \frac{x+y}{\sqrt{x^2-y}} \text{ is a real number}\}$$

For the domain of the function $x^2 - y > 0$, i.e., $y < x^2$. [$y = f(x)$]

$$\text{Domain} = \{(x, y) : y < x^2\}$$



Exercise: 4 (a) Consider the function $f(x, y) = \sin\left(\frac{x+y}{\sqrt[9]{x^2-y}}\right)$

$$D = \{(x, y) \in \mathbb{R}^2 : y \neq x^2\}.$$

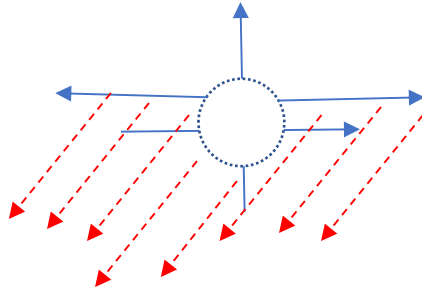
(b) Consider the function $f(x, y) = \sin(\sqrt[9]{x^2-y})$; Domain = The whole xy -plane.

Exercise: 5a Consider the function $f(x, y) = \sqrt[3]{x+1} - e^{\sqrt{2-y}} + \ln \sqrt{x^2 + y^2 - 9}$.

Solution: $g(x, y) = \sqrt[3]{x+1} = \frac{\sqrt[3]{x+1}}{1}$; $D = \text{All real numbers.}$

$$h(x, y) = e^{\sqrt{2-y}} \text{ ; } D = \{(x, y): 2 - y \geq 0\} = \{(x, y): y \leq 2\}$$

$$k(x, y) = \ln \sqrt{x^2 + y^2 - 9} \text{ ; } D: x^2 + y^2 > 9$$



Exercise: 5b Consider the function $f(x, y) = \sqrt[4]{2-x} - e^{\sqrt[3]{2-y}} + \ln \sqrt{x^2 + y^2 - 9}$.

Find the domain and sketch it.

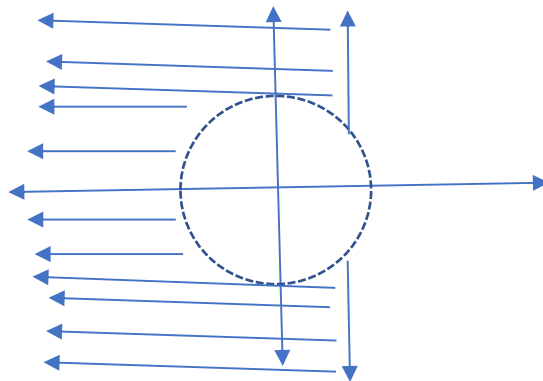
Solution: For the function $z = \sqrt[3]{2-y}$, z can be anything

For the function $w = e^{\sqrt[3]{2-y}}$, w can be anything. Hence the exponential function does not change the domain of the function f .

For the domain we get, $2 - x \geq 0$ and $x^2 + y^2 - 9 > 0$, that is,

$$x \leq 2 \text{ and } x^2 + y^2 > 3^2$$

Domain = $\{(x, y) | x \leq 2 \text{ and } x^2 + y^2 > 3^2\}$.



The domain is the region that is outside the circle $x^2 + y^2 = 3^2$ and to the left of the vertical line $x = 2$.

Exercise: 6 Consider the function $f(x, y) = \sqrt[3]{x+1} - \sqrt[4]{1-y} + \ln \sqrt{x^2 + y^2 - 4}$.

Solution: $z = \sqrt[3]{x+1}$ $D = \text{All real numbers}$,

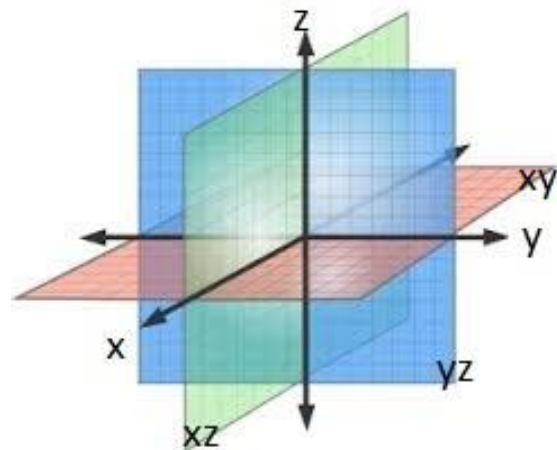
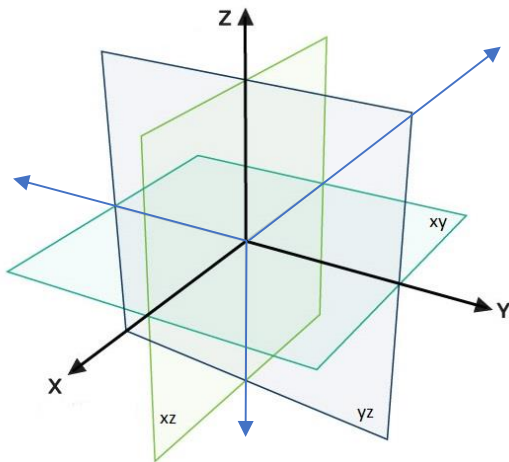
$z = \sqrt[4]{1-y}$, $D = 1 - y \geq 0$, that is $y \leq 1$

$z = \ln \sqrt{x^2 + y^2 - 4}$, $D = x^2 + y^2 - 4 > 0$, $x^2 + y^2 > 2^2$

Acknowledgement: Images are taken from internets. Thanks to owners who have created these beautiful images.

- 1) Graphing Functions
- 2) Level Curves

3D –Coordinates: xyz –Space

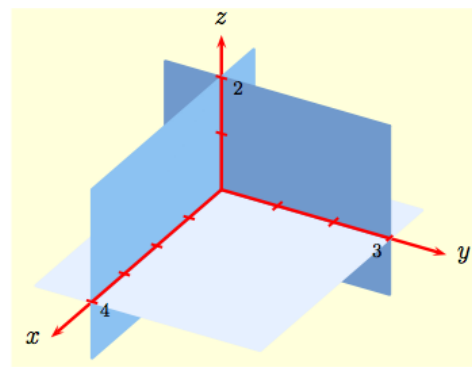
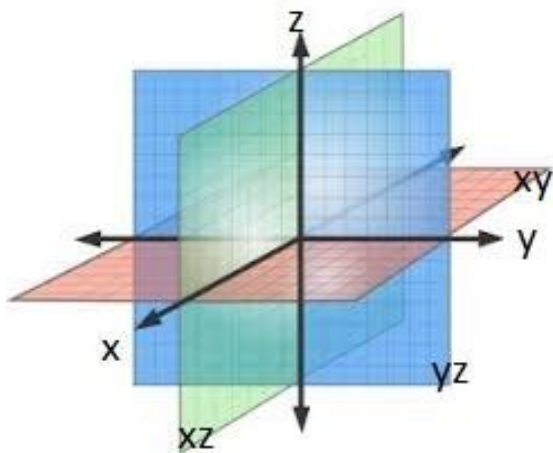
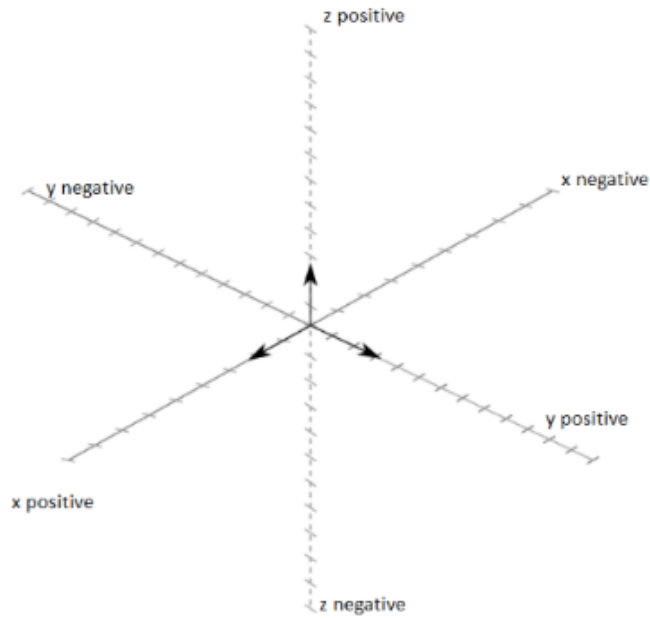


For a point (x, y, z) ,

x is the signed distance from the yz –plane

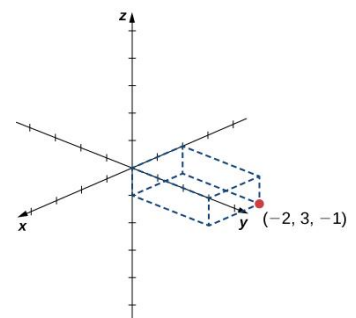
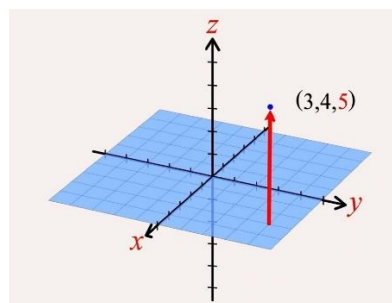
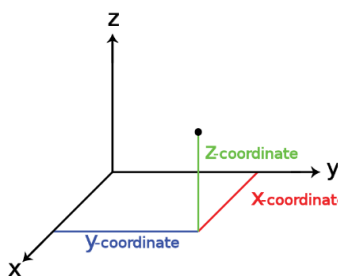
y is the signed distance from the xz –plane

z is the signed distance from the xy –plane



One Viewpoint: $x, y, z > 0$

There are **eight octants** in xyz -space. In the first octant, $x, y, z \geq 0$.



$$(x, y, z) \rightarrow ((x, y), z)$$

Coordinates of the origin $(0, 0, 0)$.

$x = 0$ on the yz –plane. So, the equation of the yz –plane is $x = 0$.

$y = 0$ on the xz –plane. So, the equation of the xz –plane is $y = 0$.

$z = 0$ on the xy –plane. So, the equation of the xy –plane is $z = 0$.

On the x –axis, coordinates of a point is given by $(x, 0, 0)$.

On the y –axis, coordinates of a point is given by $(0, y, 0)$.

On the z –axis, coordinates of a point is given by $(0, 0, z)$.

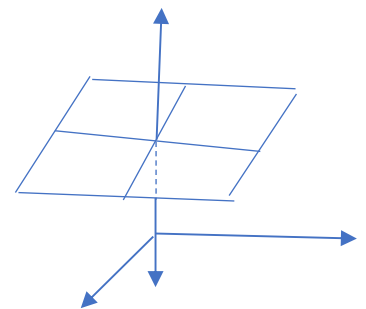
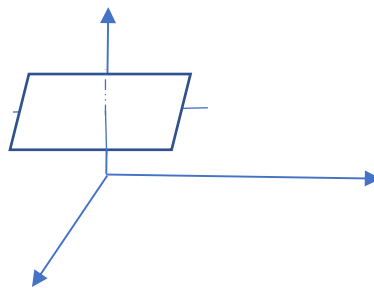
Note: In the 2D, $y = c$, c –constant $\rightarrow$ is a horizontal line, perpendicular to the y –axis at the point $(0, c)$ and parallel to the x –axis.

Graphs of the function $z = f(x, y)$:

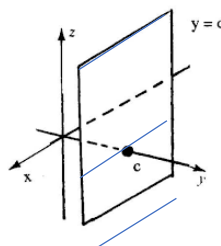
Graphs of Planes:

1) Planes parallel to coordinate planes

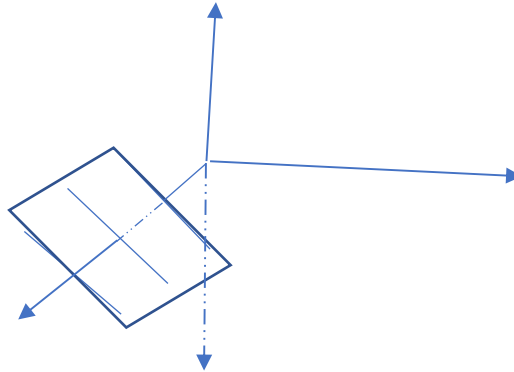
- a. The plane $z = c$ is perpendicular to the z –axis at the point $(0, 0, c)$ and is parallel to the xy –plane.



- b. The plane $y = c$ is perpendicular to the y –axis at the point $(0, c, 0)$ and is parallel to the xz –plane.



- c. The plane $x = c$ is perpendicular to the x -axis at the point $(c, 0, 0)$ and is parallel to the yz -plane.



2) Graph of the plane $x + y + z = 1$.

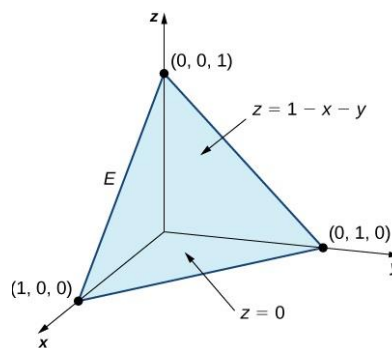
➔ Portion of the plane in the First Octant.

Solution: Given, $x + y + z = 1$. That is, $z = 1 - x - y$, i.e., $z = f(x, y) = 1 - x - y$.

x -intercept: Point on the x -axis, $y = 0, z = 0$. So, the point on the x -axis is given by $(1, 0, 0)$.

y -intercept: Point on the y -axis, $x = 0, z = 0$. So, the point on the y -axis is given by $(0, 1, 0)$.

z -intercept: Point on the z -axis, $x = 0, y = 0$. So, the point on the z -axis is given by $(0, 0, 1)$.



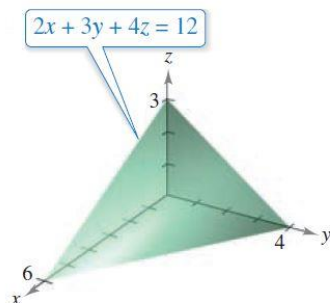
Given, $x + y + z = 1$. That is, $z = 1 - x - y$, that is, $z = f(x, y) = 1 - x - y$.

On the xy -plane: $z = 0$. We get $x + y = 1$.

On the yz -plane: $x = 0$. We get $y + z = 1$.

On the xz -plane: $y = 0$. We get $x + z = 1$.

3) Graph of the plane $2x + 3y + 4z = 12$. Homework



Definition: For any real number x , $\sqrt{x^2} = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

$\sqrt{x^2} = x$ only for $x \geq 0$. $\sqrt{9} = 3$ and $-\sqrt{9} = -3$.

This means, if the real number y is given by $y = \pm\sqrt{x}$, then by the equation $y = \sqrt{x}$ we get only non-negative real values of y . This is called the positive square root. The negative square root is given by $y = -\sqrt{x}$ which gives only non-positive real values.

[$y = x^2$ is one - to - one $(-\infty, 0]$, Inverse function: $y = -\sqrt{x}$

and $y = x^2$ is one - to - one $[0, \infty)$], Inverse function: $y = \sqrt{x}$

Note that $\sqrt{4} = \sqrt{(-2)^2} = |\pm 2| = 2$ and $-\sqrt{4} = -2$.

$$y = -|x| = \begin{cases} -(-x), & x < 0 \\ -(x), & x \geq 0 \end{cases} = \begin{cases} x, & x < 0 \\ -x, & x \geq 0 \end{cases}$$

Solve the equation $x^2 = 4 \Rightarrow \sqrt{x^2} = \pm \sqrt{4} = \pm 2$.

$\sqrt{x^2} = |x| = \sqrt{4} \Rightarrow |x| = 2$, that is, $x = \pm \sqrt{4} = \pm 2$.

Note: If $x^2 = 9$, then $x = \pm \sqrt{9} = \pm 3$. Here, $x = \sqrt{9} = 3$, $x = -\sqrt{9} = -3$

4) Graph the function $f(x, y) = \sqrt{x^2 + y^2}$.

Solution: Set $z = f(x, y) = \sqrt{x^2 + y^2}$. So, $z = \sqrt{x^2 + y^2}$. Hence, $z \geq 0$.

On the xy -plane: $z = 0 \Rightarrow \sqrt{x^2 + y^2} = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow x = y = 0$.

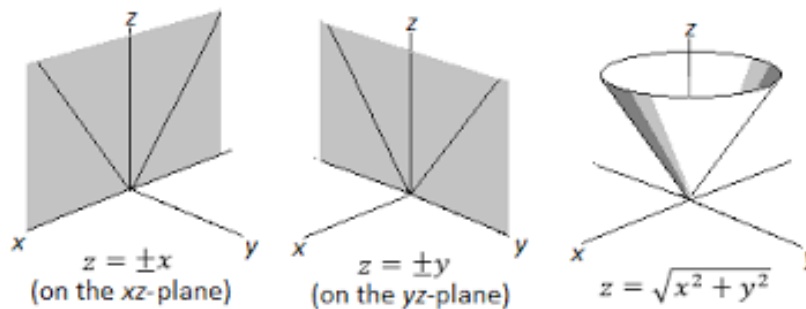
Hence the only point on the xy -plane is $(0, 0, 0)$.

On the yz -plane: $x = 0 \Rightarrow z = \sqrt{0 + y^2} = \sqrt{y^2} \Rightarrow z = |y| = \begin{cases} -y, & y < 0 \\ y, & y \geq 0 \end{cases}$

[Recall: $y = |x|$, V - shaped graph, opening upward on the xy - plane,

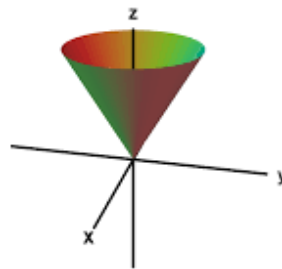
symmetric about the y - axis, y - output, Vertex = $(0, 0)$]

On the xz -plane: $y = 0 \Rightarrow z = \sqrt{x^2 + 0} \Rightarrow z = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$



where $z \geq 0$.

The graph of the function $z = f(x, y) = \sqrt{x^2 + y^2}$



Homework:

We know the graph of the function $z = f(x, y) = \sqrt{x^2 + y^2}$

- 5) Graph the function $z = f(x, y) = -\sqrt{x^2 + y^2} \rightarrow$ Reflection about the xy -plane
- 6) Graph the function $z = f(x, y) = 4 - \sqrt{x^2 + y^2} = -\sqrt{x^2 + y^2} + 4 \rightarrow$ Reflection about the xy -plane and shifting 4 units upwards

7) Graph the function $f(x,y) = \sqrt{x^2 + y^2} + 1$. shifting 1 unit upwards

8) Please submit: Graph the function $f(x,y) = -\sqrt{x^2 + y^2} - 1$

Note: If we know the graph of $y = f(x)$, then

(i) we get $y = -f(x)$ by taking the reflection about the x -axis

(ii) we get $y = f(x) + 2$ by shifting the graph upwards by 2 units

we get $y = f(x) - 2$ by shifting the graph downwards by 2 units

$[y = |x|$ and $y = -|x|$ are x -axis reflection of one another.

Similarly, $z = \sqrt{x^2 + y^2}$ and $z = -\sqrt{x^2 + y^2}$ are xy -plane reflection of one another]

9) Graph the function $f(x,y) = x^2 + y^2$.

Solution: Set $z = f(x,y) = x^2 + y^2$. So, $z = x^2 + y^2$. Hence, $z \geq 0$.

On the xy -plane: $z = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow x = y = 0$.

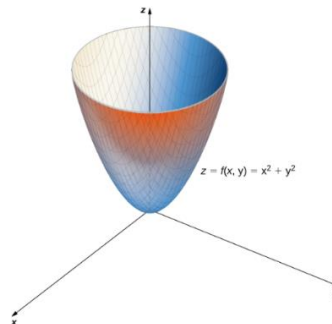
Hence the only point on the xy -plane is $(0, 0, 0)$.

On the yz -plane: $x = 0 \Rightarrow z = y^2$

[Recall: $y = x^2$, U - shaped graph, opening upward on the xy - plane, symmetric about the y - axis, y - output, Vertex = $(0,0)$]

On the xz -plane: $y = 0 \Rightarrow z = x^2$;

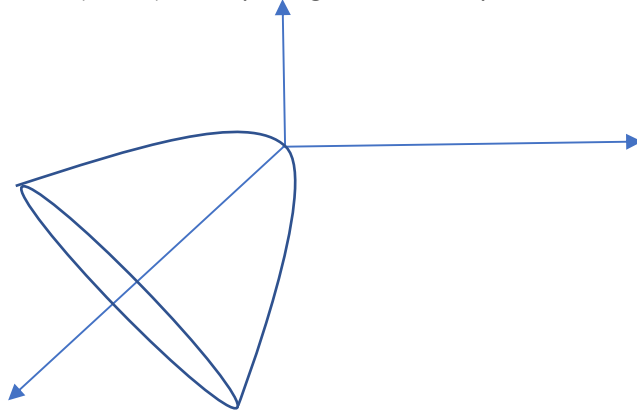
U - shaped graph, opening upward on the xz - plane, symmetric about the z - axis, Vertex = $(0,0)$



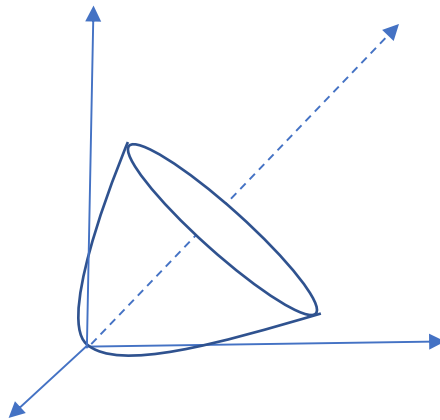
The graph is a paraboloid $z = x^2 + y^2$.

[Exercise 9 is done here]

Note that the graph is a paraboloid $x = f(y, z) = y^2 + z^2$. It's a paraboloid, symmetric about the x -axis, with vertex at the $(0, 0, 0)$ and opening towards the positive x -axis.



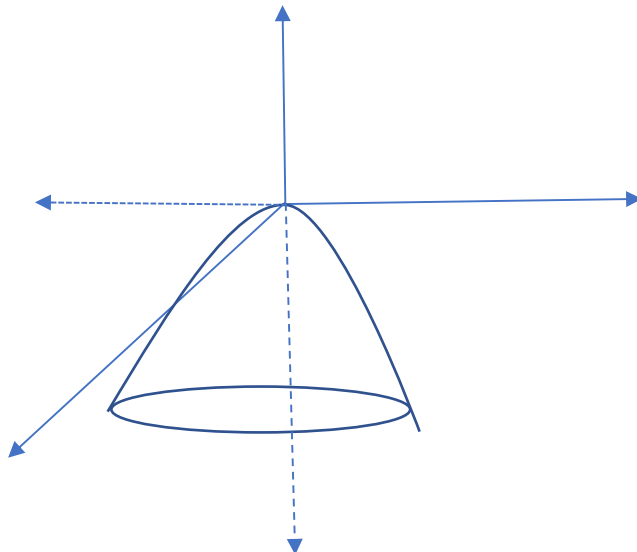
The graph of $x = -(y^2 + z^2)$ is the reflection of the graph of $f(y, z) = y^2 + z^2$ about the yz -plane



Homework:

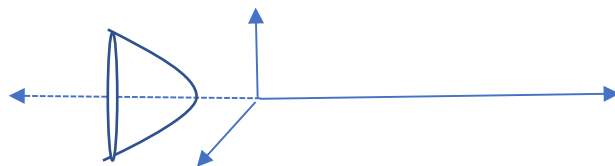
10) Graph the function $z = -(x^2 + y^2)$.

The graph $z = -(x^2 + y^2)$ is the reflection of the graph of $f(x, y) = x^2 + y^2$ about the xy -plane.



If we know the graph of $z = x^2 + y^2$, then

- 11) Graph the function $f(x, y) = 4 - (x^2 + y^2)$.
- 12) Graph the function $f(x, y) = x^2 + y^2 + 1$.
- 13) Graph the function $f(x, y) = -[(x^2 + y^2) + 3]$.
- 14) Graph the function $f(x, z) = -[(x^2 + z^2) + 3]$.



Note that the graph of $z_2 = 4(x^2 + y^2)$ is vertically stretched of the graph $z_1 = x^2 + y^2$.

Sphere: $x^2 + y^2 + z^2 = \rho^2$.

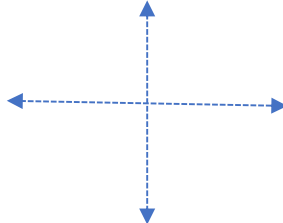
Ellipsoid: $ax^2 + by^2 + cz^2 = \rho^2$, $a, b, c > 0$ and $a \neq b \neq c$.

Q and A Session: 1

1. Sketch the domain of $f(x, y) = \ln(xy)$.

Solution: For the domain of the function, set $xy > 0$, that is,

$$(y > 0 \text{ and } x > 0) \text{ OR } (y < 0 \text{ and } x < 0)$$



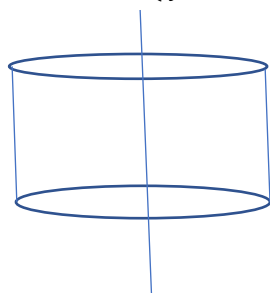
Sketch of D=??

2. Sketch the domain of $f(x, y) = \sqrt{xy}$.

Solution: For the domain of the function, set $xy \geq 0$, that is,

$$(y \geq 0 \text{ and } x \geq 0) \text{ OR } (y \leq 0 \text{ and } x \leq 0)$$

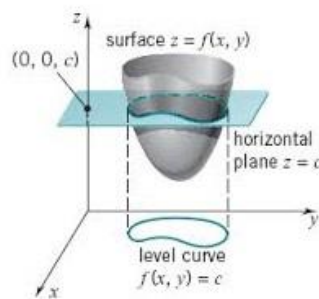
Sketch of D=??



Elliptic Cylinder, base and height of this cylinder is an ellipse.

Definition: Level Curves

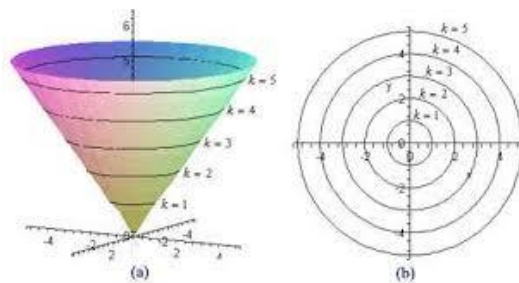
If the surface $z = f(x, y)$ is cut by the horizontal plane $z = k$, then at all the points on the intersection we have $f(x, y) = k$. **The projection of this intersection onto the xy –plane is called the level curve of height k or the level curve with constant k .** A set of level curves of $z = f(x, y)$ is called a contour plot or contour map of f .



For example, for the function $z = f(x, y) = \sqrt{x^2 + y^2}$ the level curves for $z = k$, $k = 0, 1, 2, 3, 4, 5$ are given by the graph

Defined by $f(x, y) = \sqrt{x^2 + y^2} = k$, that is, $\sqrt{x^2 + y^2} = k$, $k = 0, 1, 2, 3, 4, 5$. That is,

$$x^2 + y^2 = k^2, \quad k = 0, 1, 2, 3, 4, 5.$$



Equation $\sqrt{x^2 + y^2} = k$; $k = 0, 1, 2, 3, 4, 5$

$$k = 0: \sqrt{x^2 + y^2} = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow (0, 0)$$

$$k = 1: \sqrt{x^2 + y^2} = 1 \Rightarrow x^2 + y^2 = 1^2$$

$$k = 5: \sqrt{x^2 + y^2} = 5 \Rightarrow x^2 + y^2 = 5^2$$

Definition: Level Surface

If the surface $w = f(x, y, z)$ is cut by the equation $w = k$, then at all the points on the intersection we have $f(x, y, z) = k$. The projection of this intersection onto the xyz -space is called the level surface with constant k .

For Example, for $w = x^2 + y^2 - z$ with $w = 1$: Level surface: $1 = x^2 + y^2 - z$,

that is, $z = x^2 + y^2 - 1$; so the level surface is a paraboloid opening upward with vertex at the point $(0, 0, -1)$.

13*) Graph the function $f(x, y) = \sqrt{x^2 + y^2 - 1}$. → Please read this, and try to understand.

Set $z = f(x, y) = \sqrt{x^2 + y^2 - 1}$. So, $z = \sqrt{x^2 + y^2 - 1}$. Hence, $z \geq 0$.

Domain: $\{(x, y): x^2 + y^2 \geq 1\}$, Range: $\{z : z \geq 0\}$.

Understanding: Set $z = k$, $k = 0, 1, 2, 3, 6$,

Level curves: $z = \sqrt{x^2 + y^2 - 1}$, $z = k$, $k = 0, 1, 2, 3$,

Now, $\sqrt{x^2 + y^2 - 1} = k$, $k = 0, 1, 2, 3$,

$k = 0$: $x^2 + y^2 = 1$, circle of radius 1 with center at $(0, 0)$. Height of the level curve $k = 0$.

$k = 1$: $x^2 + y^2 = 2$, circle of radius $\sqrt{2}$ with center at $(0, 0)$. Height of the level curve $k = 1$.

$k = 2$: $x^2 + y^2 = 5$, circle of radius $\sqrt{5}$ with center at $(0, 0)$. Height of the level curve $k = 2$.

$k = 3$: $x^2 + y^2 = 10$, circle of radius $\sqrt{10}$ with center at $(0, 0)$. Height of the level curve $k = 3$.

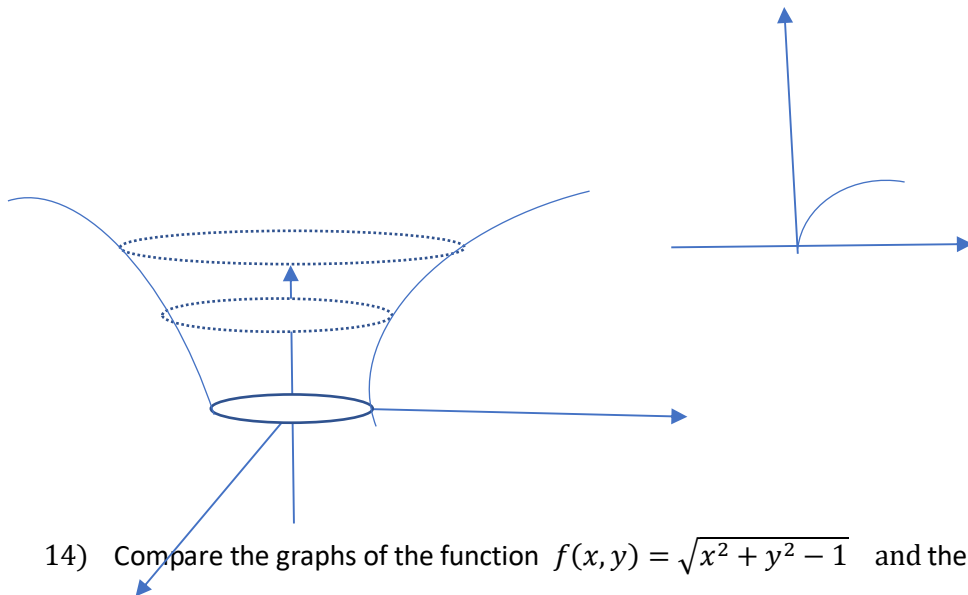
$$z = \sqrt{x^2 + y^2 - 1}$$

Notice that on the xy - plane: $z = 0 \rightarrow 0 = \sqrt{x^2 + y^2 - 1}$, that is, $x^2 + y^2 = 1$.

Notice that on the yz - plane: $x = 0 \rightarrow z = \sqrt{y^2 - 1}$.

Notice that on the xz - plane: $y = 0 \rightarrow z = \sqrt{x^2 - 1}$.

Recall: $y = \sqrt{x}$, y - output axis. This is a graph that is concave downward.



- 14) Compare the graphs of the function $f(x, y) = \sqrt{x^2 + y^2 - 1}$ and the function $f(x, y) = \sqrt{x^2 + y^2 + 1}$.

$$f(x, y) = \sqrt{x^2 + y^2 + 1} : D = xy\text{-plane}, R = z \geq 1.$$

Hint: You need to understand the level curves to see the graph of $f(x, y) = \sqrt{x^2 + y^2 - 1}$